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Capability Surprise and the Pricing of an Early-Commercial-Stage Foundation-Model Lab: Evidence from Zhipu AI (2513.HK)

Abstract

In January 2026, Zhipu AI (Knowledge Atlas Technology, 2513.HK) became the first foundation-model laboratory to list publicly anywhere, under Hong Kong's Chapter 18C listing regime. By 26 June 2026, its shares had risen about seventeen-fold to a market capitalisation near US$114bn—an equity value roughly 570 times its ∼US$200m of expected 2026 revenue—despite deep losses, negative book equity and cloud-deployment margin pressure. We ask what such a firm is worth and how its price forms. A three-scenario DCF, a reverse DCF, a comparables check and a real-options reading place fundamental value at only HK$12–82 per share (about 1.6% of the market price); the price instead embeds a winner-take-all outcome of more than US$100bn revenue by 2035. Because conventional earnings-surprise measures are economically uninformative for a deeply loss-making early-stage lab, we replace them with *capability-surprise* events (model releases and benchmark jumps). The market prices capability quickly and discriminately—separating Zhipu from peers on model quality, not sector—with case-study evidence consistent with post-capability-announcement drift. The valuation gap is best read as capability momentum priced as an option, not a simple bubble.

Keywords: foundation models; Chapter 18C; pre-profit valuation; real options; market efficiency; capability surprise; PEAD/PCAD.

1 Introduction

The first half of 2026 produced an event with no precedent in capital-market history: within forty-eight hours, two Chinese "AI tiger" laboratories—Zhipu AI (2513.HK, 8 January) and MiniMax (00100.HK, 9 January)—became the first foundation-model developers to go public anywhere, listing on the Hong Kong Stock Exchange. Zhipu listed under Chapter 18C, the regime for specialist-technology firms that admits pre-profit issuers. By 26 June 2026, its shares had compounded roughly seventeen-fold to a market capitalisation near US$114bn on FY2026 expected revenue of about US$200m—an implied equity-value-to-revenue multiple near 570×[1,2]. The broader frontier-AI financing wave has therefore moved from private markets into public price discovery.

This paper poses three questions: **(i)** what is Zhipu worth on fundamentals; **(ii)** what does the re-rating imply—efficient pricing of capability, exuberance, or option value; and **(iii)** is the open-weight, cost-disruption strategy sustainable? Our contribution is methodological: because an early-stage lab has no earnings, the standard earnings-surprise event study[3,4] has nothing to bite on, so we reframe the information event as a *capability surprise*—a model release or benchmark jump—adapting the surprise-event (*chao yuqi*) design of Guosen Securities[5] to a new asset class. The weight of the paper falls on the valuation (Section 4) and the event study (Section 5); Sections 2–3 give only the context they require.

2 Background: Company, Financials, and Listing

Business model. Founded in 2019 out of Tsinghua University's Knowledge Engineering Group, Zhipu develops the GLM ("General Language Model") family and monetises through usage-based API access, enterprise subscriptions, and private/on-premise deployment. Its strategic choice is an *open-weight funnel*: flagship models are released with open weights and aggressive token pricing, converting developer mindshare into a usage flywheel via the Z.ai and Z.code platforms.

Financials (full statements in Appendix A). Three facts dominate. *(1)* Revenue is explosive but tiny: RMB 57.4m (2022) → RMB 312.4m (2024), with H1 2025 revenue of RMB 190.9m. For the DCF we use a FY2025E base of about RMB 724m (US$102m), a run-rate estimate rather than a reported full-year figure. *(2)* Total gross margin remained positive (50.0% in H1 2025), but the strategically important cloud-deployment margin fell from 76.1% (2022) to 3.4% (2024) and to a 0.4% gross loss in H1 2025 as API prices were cut. *(3)* Losses and leverage are deepening: H1 2025 net loss was RMB 2.36bn, and at 30 June 2025 the group reported net liabilities of RMB 6.15bn[2]. Zhipu is therefore *early-commercial-stage and pre-profit*, with negative book equity—so ratio analysis is of limited use and the informative metrics are growth, cloud monetisation, and cash runway (Figure 1).

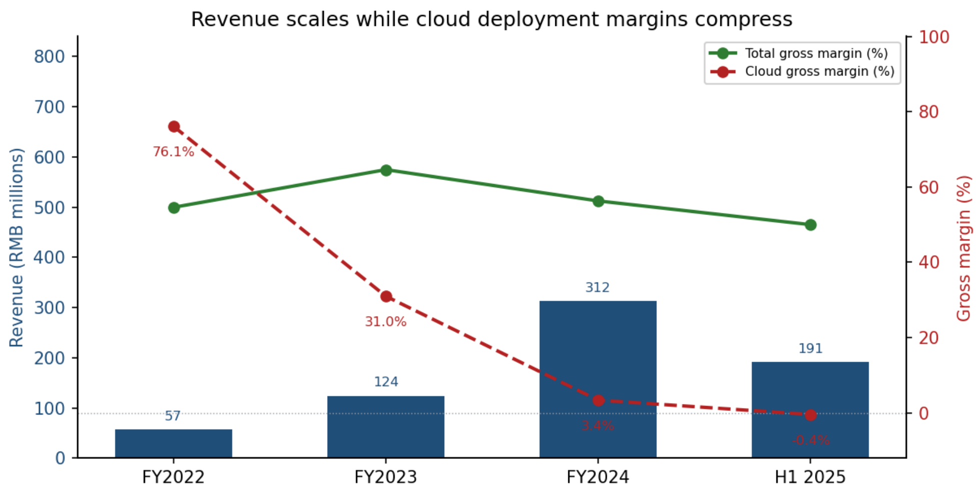

Figure 1: Zhipu's financial profile. Revenue compounds rapidly (RMB millions, left axis). Total gross margin remains positive, but cloud deployment margin compresses from 76.1% in 2022 to a 0.4% gross loss in H1 2025 as low-priced, compute-heavy API usage buys share. Source: Chapter 18C prospectus[2].

Listing and market reaction. Zhipu priced at HK$116.20, raising about HK$4.3bn with HK$3bn of cornerstone commitments, at an implied valuation near US$6.7–7.4bn[2]. The listing is strategically remarkable on three counts: Chapter 18C gave a loss-making lab permanent public capital years before profitability; its first-listed-foundation-model status conferred a certification premium; and the stock then re-rated to about HK$2,046 by 26 June 2026 (≈ +1,660% versus IPO; realised annualised volatility near 193%; Figure 7). MiniMax, listing one day later at HK$165, doubled on debut and then *de-rated* by roughly 60% from its peak—a divergence we exploit in Section 5.

3 Industry, Macro, and Strategy

Structure and strategy. In Porter terms the foundation-model layer is harsh: supplier power (advanced compute) is high and politically rationed, buyer switching costs are low because open weights commoditise the layer, and the substitutes are rival open-weight models. Zhipu's answer is leaderboard dominance through *cost disruption*: a mixture-of-experts GLM-5 line (GLM-4.6: ≈355B parameters, 32B active, 200K context) that topped coding benchmarks—GLM-5.1 ranked #1 on SWE-Bench Pro at 58.4%, and GLM-5.2 (June 2026) shipped with MIT open weights and a one-million-token context—priced below closed competitors, now paired with nascent pricing power (8–17% API price rises with each GLM-5.x release)[6]. This cadence defines the event taxonomy of Section 5 (Figure 2). The main weakness is self-inflicted: the same cadence commoditises its own product, while cloud-margin pressure and compute dependence cap monetisation.

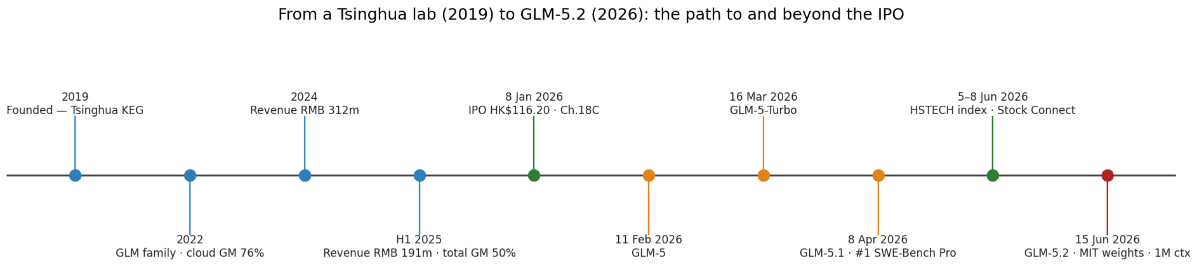

Figure 2: Company timeline from the 2019 founding (Tsinghua Knowledge Engineering Group) through the 2026 Chapter 18C listing and the GLM-5.x release cadence. Each open-weight, cost-disruptive launch is the unit of *capability surprise* studied in Section 5 and the primary currency of value for an early-commercial-stage lab; milestones are evenly spaced for legibility. Sources: Chapter 18C prospectus[2] and official model cards and technical documentation[6].

The Hong Kong AI-IPO landscape. The 2026 listing wave extends beyond Zhipu and MiniMax. Among Hong Kong–listed AI firms, SenseTime (0020.HK) has pivoted toward generative AI via its SenseNova models, Fourth Paradigm (6682.HK) provides enterprise ML infrastructure, and Wenge AI (01956.HK, listed 26 June 2026, +84% on debut) focuses on decision intelligence[7]—none is a direct frontier-model comparable. In the upcoming IPO pipeline, Moonshot AI (Kimi) is reportedly preparing a Hong Kong listing, and other Chinese frontier labs (StepStar, 01.AI) may similarly seek public capital. These prospective listings reinforce the first-mover premium Zhipu's Chapter 18C listing confers. *For this paper's valuation, only MiniMax is a direct comparable* (Section 4.4).

Macro and flows. Chinese AI is a policy priority (state/SOE demand); U.S. compute-export controls force the architectural efficiency that doubles as Zhipu's cost edge; and Hong Kong's market structure sets the marginal buyer—Zhipu joined the Hang Seng Tech Index (5 June) and Southbound Stock Connect (8 June 2026), drawing about HK$920m of net southbound buying on its first eligible day[8]. These flow events matter for Section 5, where we separate them from capability events.

4 Valuation

4.1 Cost of capital

We estimate the cost of equity with the CAPM. Given Zhipu's extremely short post-IPO trading history, an estimated regression beta would be unstable and dominated by IPO- and event-driven shocks; we therefore construct a bottom-up beta from global AI-software comparables, giving $\beta \approx 1.6$. With a risk-free rate of 4.0% and an equity risk premium of 6.0%, $K_e = 4.0\% + 1.6 \times 6.0\% = 13.6\%$; as the firm is effectively all-equity, WACC $\approx 13.5\%$. Throughout we use HK$7.8/US$, RMB 7.1/US$, $\approx$435m shares, and net cash of about **US$0.55bn**—net IPO proceeds of ~HK$4.3bn (~US$0.55bn) plus pre-IPO cash of ~US$0.03bn. The pre-IPO net-liability position is dominated by redeemable preferred shares that convert to ordinary equity on listing, so it is not post-IPO debt; treating any residual as debt would only lower equity value and reinforce our conclusion. Appendix C lists the comparable set and unlevering bridge.

Two caveats, deliberately retained. We keep this single comparables-based WACC for transparency, but two simplifications deserve flagging. *(i) The beta source.* The bottom-up β

is built from mature global AI-software names (Salesforce-, Palantir-type firms) with healthy margins and conventional cash-flow profiles, whereas Zhipu today behaves more like a high-capex frontier "hard-tech" lab; a β drawn from pre-profit infrastructure or biotech comparables—asset classes with the same binary payoff structure—would be higher, so the software-based WACC is generous to valuation. *(ii) A static WACC.* Holding WACC fixed at 13.5% across the full 2026–2035 horizon is mechanical: under the reflexivity logic developed later in this section, a break in the high-valuation loop would push Zhipu's cost of capital up non-linearly, so a fuller model would attach a dynamic risk premium or an explicit distress/default probability to the discount rate. We do not do so here because both refinements push value in the *same* direction—a higher or state-dependent WACC lowers the DCF—and the sensitivity analysis (Figure 5) already shows that even a markedly *lower* WACC cannot close the gap to the market price. The binary tail that these adjustments would proxy is, in any case, handled more naturally by the real-options reading below.

4.2 Scenario DCF

We project FCFF over 2026–2035 from a US$200m base, with revenue growth fading from a scenario-specific 2027 start to a mature rate, operating margin ramping from −30% to a terminal level, a 15% tax rate, and reinvestment via a sales-to-capital ratio of 1.6–2.0; terminal value uses $g = 4\%$. When EBIT is negative, cash tax is zero and losses accumulate in a simple NOL carryforward account; positive EBIT is taxed only after the NOL balance has been absorbed. The model is built as a live spreadsheet, while the full Base-case chain is reported in Appendix B; headline results appear in Table 1.

Table 1: Three-scenario DCF (USD unless noted; per-share in HKD). Market row at 2026-06-26.

Scenario	Rev. CAGR '26–35	Term. margin	Equity value	Per share (HK$)
Bear ($p = 0.35$)	21%	18%	US$0.7B	12
Base ($p = 0.45$)	31%	28%	US$1.6B	28
Bull ($p = 0.20$)	46%	35%	US$4.6B	82
Probability-weighted	—	—	**US$1.9B**	**33**
Market (2026-06-26)	—	—	*US$114B*	*2,046*

The probability-weighted intrinsic value is about HK$33 per share—roughly **1.6% of the market price**. No reasonable point estimate of fundamentals approaches the quote.

4.3 Reverse DCF: what the price requires

We invert the model to ask what justifies the US$114bn equity value. At WACC 13.5% and 40% terminal margin, the price requires **about US$130bn of 2035 revenue**—$\sim$105% annual growth over 2026–2035. Figure 3 grids WACC (10–20%) against terminal margin (20–50%): even the most favourable cell (10%/50%) demands $\sim$ $42bn, and 15%/40% pushes to $181bn. Thus the DCF is not a target price, but a way to state the right-tail outcome embedded in the quote.

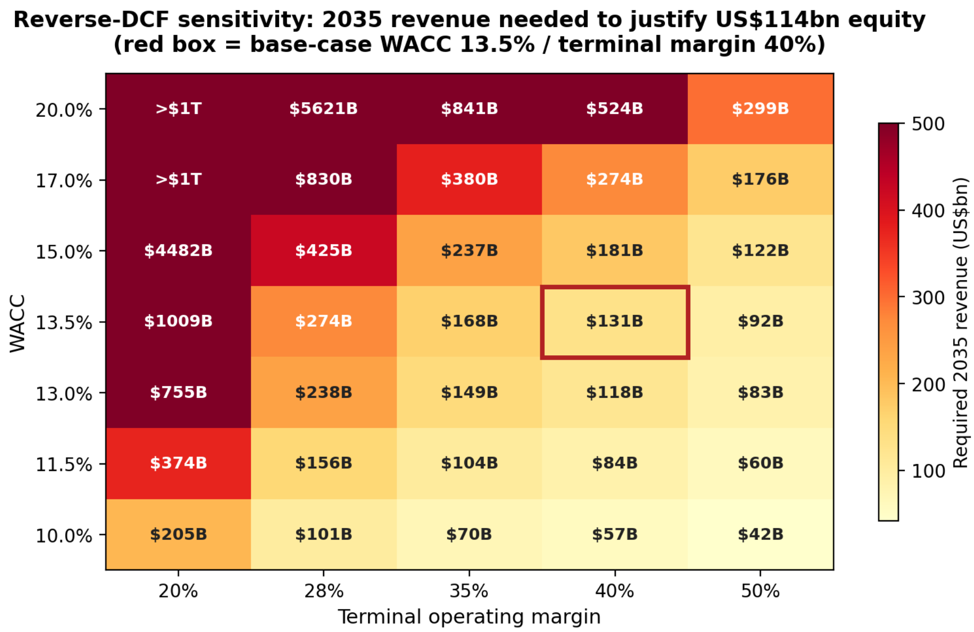

Figure 3: Reverse-DCF sensitivity: 2035 revenue (US$bn) required to justify the US$114bn equity value across WACC and terminal-margin assumptions. The red box marks the base case (13.5%/40%). Even the most favourable cell demands revenue far beyond any current AI company. Data: Table 8 in Appendix D.

4.4 Relative valuation

On FY2026E revenue Zhipu trades near **570× equity value / revenue**. For context, reported 2026 private-market reference points put OpenAI at about $852bn of value and roughly $24bn annualised revenue, and Anthropic near $965bn of value and about $47bn annualised revenue[10,11]. Those figures imply low-tens revenue multiples, not hundreds. MiniMax—the most appropriate direct comparable—trades far lower. We deliberately *exclude* Wenge AI: its government/enterprise project-delivery model is not comparable to a frontier-model-API business. Zhipu is priced roughly an order of magnitude above even the richest private peers—consistent with the reverse-DCF reading (Figure 4).

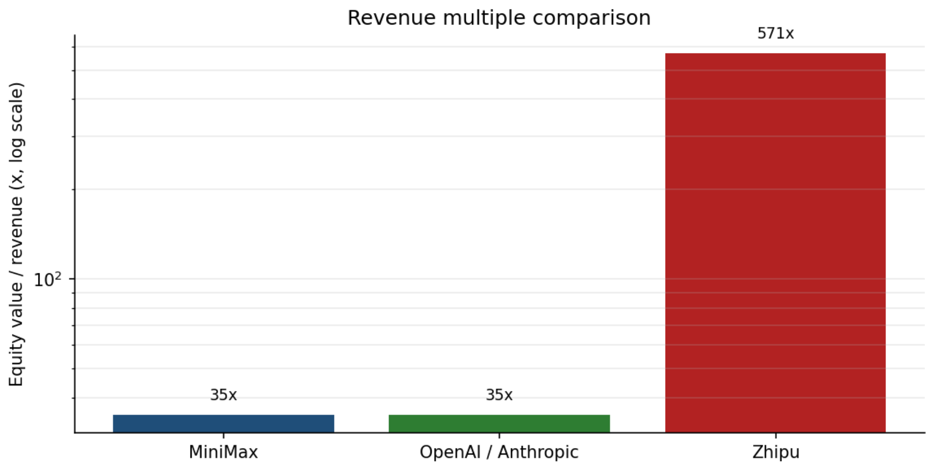

Figure 4: Equity value / revenue multiples (log scale). Zhipu's ≈570× is roughly an order of magnitude above the 20–35× at which leading private frontier labs have transacted in recent funding rounds (Section 4.4); even MiniMax sits far below. Private-lab multiples are private-market reference points, not liquid-market prices.

4.5 A real-options reading

The reconciliation is that Zhipu is best valued not as a discounted cash-flow stream but as a *call option on a winner-take-all AGI outcome*[12,13]. The DCF base case is the option's modest intrinsic value; the bulk of the market price is time value on a fat right tail—high volatility (≈193% annualised) and a long horizon both inflate that value. This separates *fundamental value* from *option value* and explains why the stock can rationally trade far above any DCF point estimate while remaining acutely sensitive to capability news—the subject of Section 5.

A practitioner lens: reflexivity. Reading the expectations embedded in the price—the reverse-DCF move—echoes the reverse-investing framework of the fund manager Feng Liu, who applies Soros's *reflexivity* to equity pricing[14]: a high valuation is not only an output but an input, as cheap equity capital funds the compute and talent that improve the models, which in turn lift the valuation. For Zhipu this loop is unusually direct—and unusually fragile, since Feng Liu also warns that firms with limitless financing needs and no self-generated cash flow (unlike a Moutai-type internal compounder) depend on the loop staying intact. If capability surprises stop arriving, the reflexive support for the option value can unwind as fast as it built.

4.6 Sensitivity and synthesis

The conclusion is not an artefact of any single assumption. Figure 5 tornadoes the Base case against its main drivers: even the most favorable one-at-a-time changes—lower WACC, a higher terminal margin, or stronger 2027 growth—raise estimated value only to the mid-HK$30s, still about two orders of magnitude below the market. Figure 6 draws the whole picture together as a valuation football field: scenario DCF (HK$12–82) and an equity-value-to-revenue cross-check (HK$72–126 at 20–35× revenue) both sit far to the left of the HK$2,046 quote, a gap of roughly 60–70×. The market is not pricing a different point on the same distribution; it is pricing a different distribution—the right-tail option of Section 4.

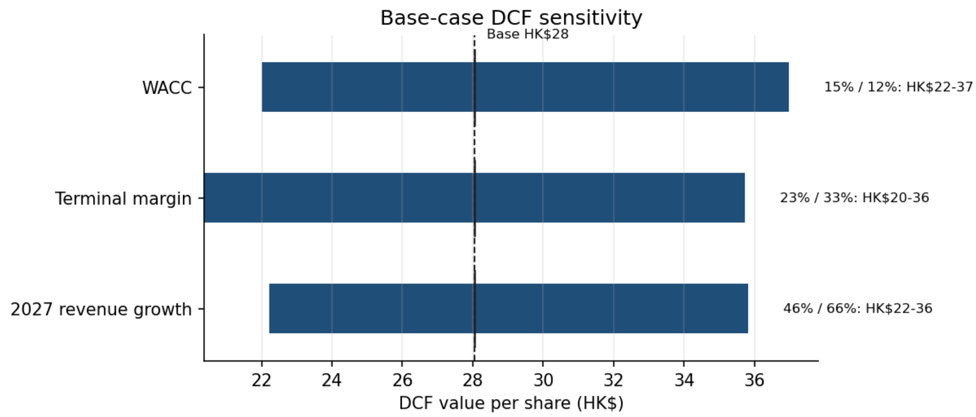

Figure 5: DCF sensitivity (Base case, per share). Bars span each driver's low/high setting around the HK$29 base; no single input moves value near the market price.

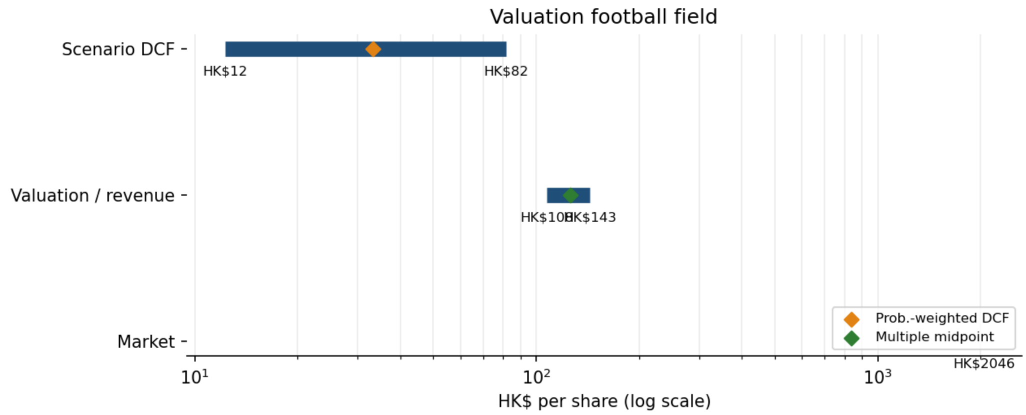

Figure 6: Valuation football field (log scale). Fundamental approaches cluster two orders of magnitude below the market price; the diamonds mark the probability-weighted DCF and the comparables midpoint.

5 Capability Surprise: An Event Study

Hypothesis. If the firm is priced on its capability trajectory, then capability information should move the price. Under semi-strong efficiency[15] the reaction should be immediate and complete; any systematic *post-capability-announcement drift* (PCAD)—the analogue of the classic post-earnings-announcement drift[3,4]—would indicate inefficiency.

Method. Event dates are identified *independently* from official model-release announcements and technical documentation[6], not from the price series. On daily prices[1] we compute mean-adjusted abnormal returns and CARs over three windows—leakage $[-5, -1]$, reaction $[0, +1]$, and drift $[+2, +10]$—following standard event-study practice[16] and the surprise-event design of Guosen Securities[5]. For each event the expected daily return is Zhipu's mean return over the pre-event estimation window $[-20, -6]$, and the abnormal return is $AR_t = R_t - \overline{R}_{[-20,-6]}$. Figure 8 then serves as a *descriptive cross-check*: several of the largest positive-return days coincide with these independently dated capability events.

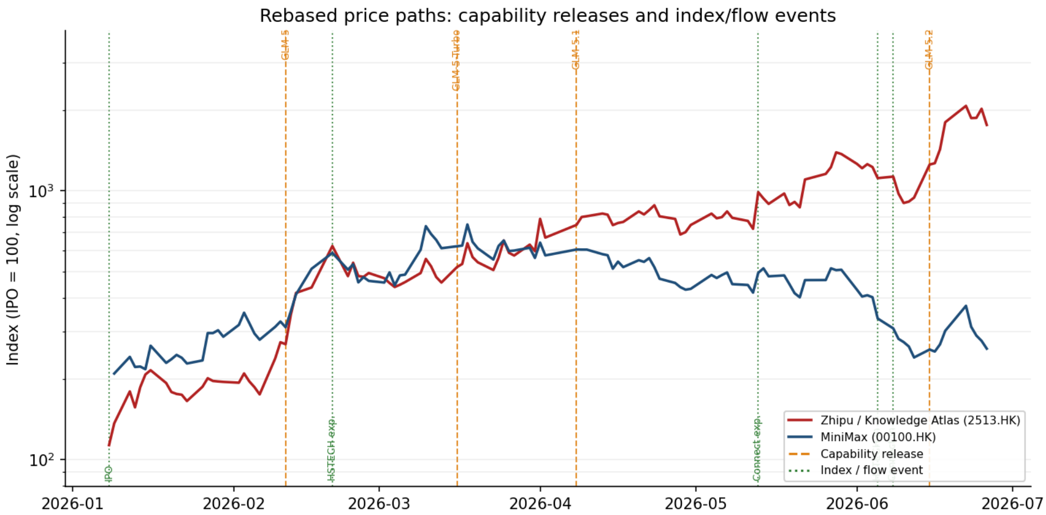

Figure 7: Rebased prices (IPO = 100, log scale, through 2026-06-26). Orange dashed lines mark capability releases (GLM-5 through GLM-5.2); green dotted lines mark index/flow catalysts (IPO, the 20 Feb and 13 May expectation spikes, and the 5–8 June Hang Seng TECH inclusion and Stock Connect access). Same sector, divergent paths: Zhipu re-rates on successive SOTA releases while MiniMax de-rates from its peak.

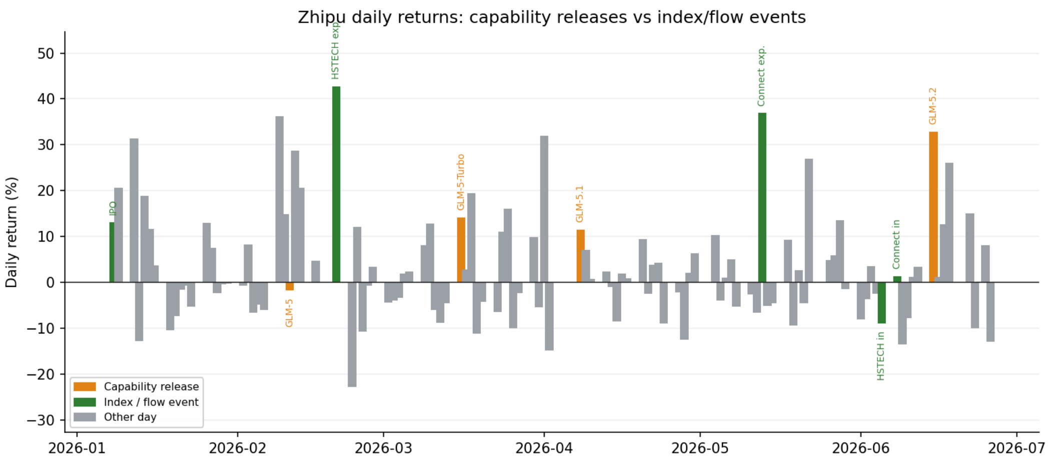

Figure 8: Descriptive cross-check: Zhipu's daily returns with capability-release days (orange) and index/flow days (green) highlighted. Several of the largest up-days coincide with independently dated GLM releases, but the two biggest—+43% on 20 Feb and +37% on 13 May—are index/flow events, not model releases. (Event dates are taken from official announcements and index/Connect schedules, not from this chart.)

Results. Averaged across the four GLM events, the reaction window is +18.7% (the market prices capability, fast) and the drift averages only +4.8% but is strongly *bimodal* (Table 2). The drift sign depends on the *type* of release: a genuine SOTA leap (GLM-5.2) continues upward, whereas incremental upgrades (GLM-5-Turbo, GLM-5.1) reverse. MiniMax's M2.7 drew a negligible reaction and a sharp subsequent de-rating (−49%, partly the unwinding of its own debut bubble), consistent with *capability surprise, not sector membership, driving the cross-section* (Figures 9–10).

Table 2: Cumulative abnormal returns around capability events (mean-adjusted, %).

Event	Day 0	Reaction $[0, +1]$	Drift $[+2, +10]$	Reading
GLM-5	2026-02-11	+22.5	+24.7	under-reaction
GLM-5-Turbo	2026-03-16	+7.7	−19.3	reversal
GLM-5.1	2026-04-08	+13.8	−14.2	over-reaction
GLM-5.2	2026-06-15	+30.8	+28.2†	strong under-reaction
MiniMax M2.7	2026-03-18	−5.5	−49.1	de-rate
Average (4 GLM)		**+18.7**	**+4.8**†	

Mean-adjusted: $AR_t = R_t - \overline{R}_{[-20,-6]}$. † GLM-5.2's $[+2, +10]$ window is *truncated* (only 7 of 9 trading days fall on/before the 2026-06-26 cut-off), so its drift—and the 4-event average drift—are provisional.

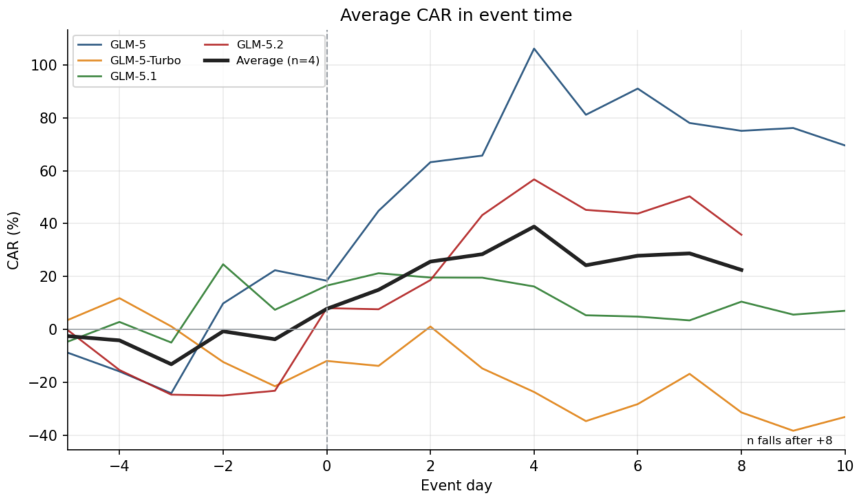

Figure 9: Average CAR in event time across the four GLM events (mean-adjusted, $[-20, -6]$ estimation window). The average is plotted only where all four events have data; the available count falls from four to three after day +8 (GLM-5.2 window truncated at the 2026-06-26 cut-off).

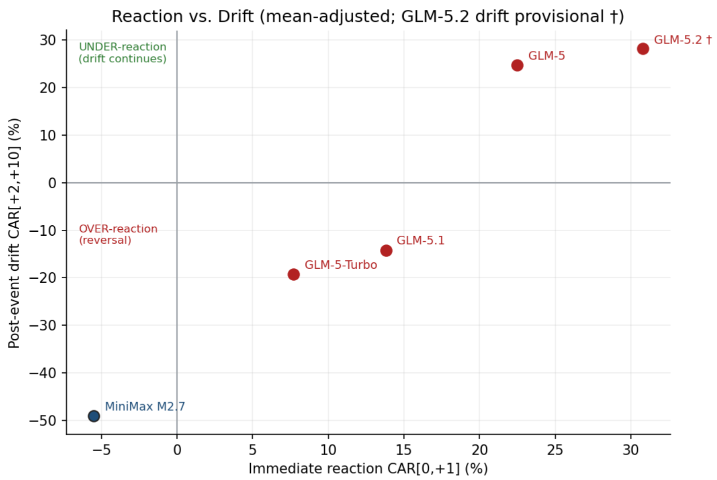

Figure 10: Reaction versus drift. SOTA leaps sit in the under-reaction quadrant; incremental upgrades in the over-reaction quadrant.

5.1 Not every spike is capability: flow events

Two of the largest blind-detected up-days do *not* map to a model release: +43% on 2026-02-20 and +37% on 2026-05-13. Both coincide with *index/flow* catalysts—rising expectations of Hang Seng Tech inclusion and Stock Connect access (realised 5 and 8 June). Separating capability events from flow events matters: the CARs above are driven by the former, while the latter explain part of the raw volatility and warn against attributing every move to model quality.

5.2 Robustness: peer-adjusted and block-bootstrap checks

Because Zhipu and MiniMax share sector and macro shocks, we re-run the study with *peer-adjusted* returns (benchmark = MiniMax's same-day return). Table 3 shows the reaction result is highly robust (+18.7% vs +18.5%), while the average drift *strengthens* (+4.8% → +19.2%): once the falling sector is removed, all four events drift positively, and GLM-5.1's apparent reversal flips to continuation (−14.2% → +12.9%). The under-reaction pattern is reinforced, not weakened, by peer adjustment.

We also maintain an expanded event catalogue as a guardrail against cherry-picking: four Zhipu capability events, one MiniMax peer event, four Zhipu index/flow events, three listing events and two candidate Wenge/Moonshot entries. Only the nine events with enough local price history enter the expanded CAR panel; excluded entries carry an explicit reason. In that panel, capability reactions remain positive, while flow events show weaker immediate reaction but strong subsequent drift, confirming why the paper separates capability from liquidity catalysts rather than pooling them mechanically.

Table 3: Robustness: mean-adjusted vs peer-adjusted CAR (%).

	Mean-adjusted		Peer-adjusted	
Event	React $[0, 1]$	Drift $[2, 10]$	React $[0, 1]$	Drift $[2, 10]$
GLM-5	+22.5	+24.7	+17.2	+13.3
GLM-5-Turbo	+7.7	−19.3	+14.6	+19.2
GLM-5.1	+13.8	−14.2	+13.4	+12.9
GLM-5.2	+30.8	+28.2†	+28.7	+31.3†
Average	+18.7	+4.8†	+18.5	+19.2†

† GLM-5.2 drift window truncated (7/9 days through 2026-06-26); the GLM-5.2 rows and the average drifts are provisional.

Block-bootstrap check. With only four events, a parametric t-test would create false precision. We therefore draw 20,000 non-event peer-adjusted return blocks with numpy.random.choice, preserving the two-day reaction and nine-/seven-day drift window lengths, and compare the actual four-event mean CAR with the bootstrap mean distribution (Figure 11). The actual peer-adjusted reaction (+18.5%) sits above the 95% null interval ($[-17.3, +14.0]$%) and at the 99.6th percentile. Drift (+19.2%) is positive but inside its wider null interval ($[-20.2, +30.6]$%), at the 78.7th percentile. Thus the block bootstrap supports a strong immediate capability reaction and only directional, not statistically conclusive, continuation.

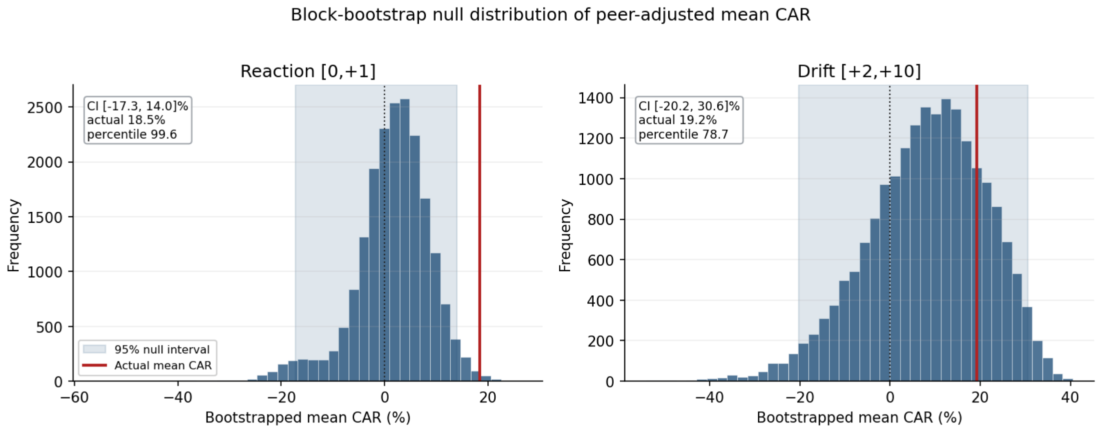

Figure 11: Block-bootstrap null distributions for peer-adjusted mean CAR. Each draw samples four non-event return blocks with replacement using numpy.random.choice; block lengths match the actual event windows. Shaded areas are percentile 95% null intervals, and red lines mark the actual four-event means.

Honesty box. This is *preliminary, diagnostic* evidence consistent with PCAD, not a proven anomaly: $n = 4$ single-firm events (plus one peer) over five months; GLM-5.2's drift window is truncated; and drift windows can overlap competing-lab news. No statistical significance is claimed—four events make an interesting undergraduate case, not a general law. The natural extensions are a wider, multi-lab event panel and an NLP-scored surprise magnitude.

6 Conclusion and Implications

A disciplined valuation cannot reconcile Zhipu's fundamentals (HK\$12–82 per share) with its HK\$2,046 market price (26 June 2026). The reverse DCF shows why: the price embeds a winner-take-all, US\$130bn-revenue outcome—a call option on frontier AI, not a cash-flow stream. The event study shows *how* that price is discovered—through capability surprises priced quickly and discriminately, with case-study evidence of under-reaction (a PCAD pattern that survives peer adjustment), while separate index/flow events drive part of the rest. The verdict is neither "efficient" nor "bubble" but **capability momentum priced as an option**. For investors, this argues for trading the capability calendar with attention to release *type*; for founders, Chapter 18C has made capability milestones the primary currency of value; for policymakers, compute access is now a direct input to capital formation. The main limitations are the small event sample and uncertain cloud monetisation.

A Financial Statements Summary

From the Chapter 18C prospectus (RMB; USD at ≈7.1).

Table 4: Income-statement summary (RMB millions unless noted).

	FY2022	FY2023	FY2024	H1 2025
Revenue	57.4	124.5	312.4	190.9
Total gross margin	54.6%	64.6%	56.3%	50.0%
Cloud gross margin	76.1%	31.0%	3.4%	−0.4%
Net loss	143.7	788.0	2,958.0	2,357.9

Table 5: Balance-sheet highlights (RMB millions, at 30 June 2025).

Item	Amount
Cash and cash equivalents	2,552.0
Net liabilities (negative equity)	−6,150.8
Net current liabilities	−7,088.9
Shares outstanding (post-IPO, approx.)	≈435m
IPO price / gross proceeds	HK$116.20 / ≈HK$4.3bn

The central operating fact is not a collapse in total gross margin, but the compression of cloud deployment economics as Zhipu cuts API prices to win share. Combined with losses and negative equity, it is why the valuation in Section 4 must rest on future-option logic rather than current fundamentals.

B Base-Case DCF Projection

Base-case chain (US$m unless noted), reproducing the live workbook: WACC = 13.5%, statutory tax = 15%, terminal $g = 4\%$, sales-to-capital = 1.8, ≈435m shares, HK$7.8/US$. Revenue grows from US$200m in 2026, with growth fading from 56% in 2027 to 8% in 2035; the 2025 base of about US$102m is a FY2025E run-rate estimate. Operating margin ramps −30%→28%. Cash tax is zero while EBIT is negative; losses accumulate as NOLs and shelter future positive EBIT until exhausted.

Table 6: Base-case projection (US$ millions unless noted).

Year	Revenue	Growth	EBIT margin	Cash tax	NOL end	NOPAT	Reinvest	FCFF
2026	200	96%	−30%	0	60	−60	54	−114
2027	312	56%	−24%	0	133	−73	62	−136
2028	468	50%	−17%	0	214	−80	87	−167
2029	674	44%	−11%	0	285	−72	114	−186
2030	930	38%	−4%	0	325	−39	142	−182
2031	1,228	32%	2%	0	297	27	165	−138
2032	1,547	26%	9%	0	163	134	177	−43
2033	1,856	20%	15%	18	0	263	172	91
2034	2,116	14%	22%	68	0	388	144	243
2035	2,285	8%	28%	96	0	544	94	450

Terminal value (2035) = $450 \times 1.04/(0.135 - 0.04)$ = US$4,925m; PV of TV = US$1,388m.
Enterprise value = US$1,015m; + net cash US$550m = equity US$1,565m = **HK$28/share**.

C Bottom-Up Beta Bridge

The bottom-up beta uses public software and AI-infrastructure comparables with observable market betas and capital structures. We unlever each beta as $\beta_U = \beta_L/[1 + (1 - T)D/E]$, using a 21% tax rate, market capitalisation as equity value, and total debt as debt[9]. Because Zhipu is effectively all-equity after IPO preference-share conversion, the relevered beta is close to the unlevered beta; we use the median bridge ($\approx$1.53) rounded to $\beta = 1.6$ to leave a modest early-stage hard-tech risk buffer.

Table 7: Comparable-company beta bridge (accessed 28 June 2026).

Comparable	Levered beta	Debt/equity	Unlevered beta
Salesforce	1.15	32.8%	0.91
Palantir	1.51	0.1%	1.51
Snowflake	1.35	3.2%	1.32
Datadog	1.55	1.5%	1.53
MongoDB	1.55	0.2%	1.55
Cloudflare	1.67	4.2%	1.62
C3.ai	2.03	4.2%	1.97
Median	—	—	**1.53**

D Reverse-DCF Sensitivity Grid

Table 8 reports the 2035 revenue (US$bn) required to justify the US$114bn equity value across a grid of WACC and terminal-margin assumptions, underlying Figure 3.

Table 8: Reverse-DCF sensitivity: required 2035 revenue (US$bn) to justify US$114bn equity.

WACC	Terminal operating margin				
	20%	28%	35%	40%	50%
10.0%	205	101	70	57	42
11.5%	374	156	104	84	60
13.0%	755	238	149	118	83
13.5%	**1,009**	**274**	**168**	**131**	**92**
15.0%	>1,000	425	237	181	122
17.0%	n/a	830	380	274	176
20.0%	n/a	>1,000	841	524	299

Bold row = base-case WACC. "n/a" = WACC $\leq g$ makes terminal value undefined at that margin. All figures assume $g = 4\%$, sales-to-capital 1.8, 15% tax, and the base-case growth path fading from 56% (2027) to 8% (2035).

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